

Exploring Natural Eye-Gaze-Based Interaction for Immersive Virtual Reality

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ABSTRACT

Eye tracking technology in a head-mounted display has undergone rapid advancement in recent years, making it possible for researchers to explore new interaction techniques using natural eye movements. This paper explores three novel eye-gaze-based interaction techniques: (1) Duo-Reticles, eye-gaze selection based on eye-gaze and inertial reticles, (2) Radial Pursuit, cluttered-object selection that takes advantage of smooth pursuit, and (3) Nod and Roll, head-gesture-based interaction based on the vestibulo-ocular reflex. In an initial user study, we compare each technique against a baseline condition in a scenario that demonstrates its strengths and weaknesses.

Index Terms: H.5.2 [Information Interfaces and Presentation]: User Interfaces—Interaction styles

1 INTRODUCTION

Eye tracking technology has been studied in the field of Human-Computer Interaction to understand the user's point of regard in analyzing user interface designs, and also as an interaction device in its own right [1]. While most prior research used eye tracking sensors for interacting with desktop monitors, recent advances in head-mounted displays (HMDs) for Virtual Reality (VR) have also driven development of head-worn eye trackers. VR HMDs with eye tracking technology are becoming more accessible, such as the FOVE HMD [2]. Using an HMD with such capability, a computer can observe and learn user attention. Well-designed eye gaze-based interaction could potentially offer more natural and implicit interaction that impacts the VR experience in a significant way.

Early investigation with eye tracking for interaction in an HMD-based VR environment has shown performance benefits compared to pointing with fingers [3]. The interaction method used was selection based on eye-fixation time which has been widely adopted for 2D interfaces to solve the Midas touch problem [1]. Fixation or dwell time is a standard delimiter for indicating a user's intention to select an object through eye gaze alone. Dwell time typically ranges from 450 ms to 1 second for novices, but can be improved over time to around 300 ms in the case of gaze typing [4]. However, this time constraint can negatively impact the user experience. For example, when the required dwell time is too short, it puts pressure on the user to look away, avoiding accidental selection, but if it is too long, it results in longer wait times.

While there are various approaches for developing novel eye-gaze-based interaction, forcing unnatural eye movements could quickly cause fatigue or eye strain. If the method is too complex, it

could end up overwhelming the user and require long training times. To prevent such problems, we need to understand natural eye movements and design interactions based on them. Prior research [5] showed four primary types of natural eye movements: (1) saccade, a quick eye movement with a fixed end target, (2) smooth pursuit, a smooth eye movement towards a moving target (3) vestibulo-ocular reflex (VOR), an automatic eye movement that counters head movement when fixating on a target, and (4) vergence, converging/diverging our eyes to look at targets at different distances. Previous research explored various interaction methods based on natural eye movements, such as detecting head gestures based on VOR [6-7], leveraging smooth pursuit for auto-calibration [8], spontaneous interaction on public displays [9], and interacting with 2D GUI controls [10-13]. However, these were mainly designed for 2D interfaces on desktop monitors or large-screen displays.

In this paper, we report on our explorations into designing novel eye-gaze-based interaction techniques leveraging natural eye movements for immersive VR experienced in an HMD. We introduce three novel interaction techniques, based on saccade, smooth pursuit, and VOR. We also report on our initial user study and discuss the relative strengths and weaknesses of the techniques.

2 NATURAL EYE-GAZE-BASED INTERACTION TECHNIQUES

Our design for eye-gaze-based interaction for immersive VR was centered on the three primary types of natural eye movements. In this section, we present three novel techniques that take advantages of different types of eye movements; (1) Duo-Reticles, inspired by saccade, (2) Radial Pursuit, using smooth pursuit, and (3) Nod and Roll, through VOR.

2.1 Duo-Reticles

Duo-Reticles (DR) is a spatio-temporal based selection technique that delivers comparable performance to gaze-dwell without the limitation of dwell time. DR waits for a confirmation from the user through reticle alignment to trigger a selection taking advantage of eye saccades, using two reticles. The first reticle, the eye-gaze reticle (ER), indicates the user's current eye-gaze location in real-



Figure 1: System setup - HTC Vive on a laptop PC and Pupil Labs eye tracker (inside HMD) ran on an iMac, streaming eye-gaze data to the laptop PC.

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time. The second reticle, the inertial reticle (IR), indicates a moving-average gaze location from the near past. Over time, the IR's position updates and moves toward the ER, but stops short at a minimal Euclidean distance threshold, α . The IR is basically playing cat and mouse with the ER; it can never quite catch it due to the restricted distance, α , that keeps them apart. Only when the ER moves to the IR can they be aligned and a selection can happen (see Figure 2A-1 to 2A-3).

We enforce the selection process with a minimum alignment distance and time. The minimum alignment distance parameter relies on the precision of the eye tracking technology and should be determined through experimentation. In terms of time, there is a natural fixation period of around 200-300 ms where the eyes stay relatively still; we take this as an appropriate minimum alignment time.

The strength of using saccade is in its incredible speed, where its velocity can reach as high as 500° per second [5]. Since the saccade velocity depends on how far our eyes move, the maximum speed occurs before the midpoint and drops until the target is reached. DR measures the current velocity of the ER, v , and applies the normalized constant, c , to find the moving average position for the IR. The IR behaves in a similar way as a saccade, since greater acceleration is applied when the saccade travels a greater distance and as the IR approaches the ER, it decelerates and comes to a complete stop.

Equation 1 gives the current position of the IR in terms of the ER's position and its velocity.

$$x_{IR}(n) = \frac{\sum_{i=n-k+1}^n x_{ER}(i)}{k}, k = \lfloor t \cdot vc \rfloor \quad (1)$$

$$\text{where } 1 \leq k \leq n-1 \text{ and } |x_{IR}(n) - x_{ER}(n)| \geq \alpha$$

The current position of the IR is defined by the average of the ER's past and present positions where n is the current frame and k is the number of frames to average. k is defined as the floored product of t , which is the maximum number of frames to buffer, v is the eye movement speed, and c is the constant weight that normalizes the eye movement speed. The value of t is determined

based on trial and error, varying due to the framerate and latency of eye tracking. The distance difference between the IR and the ER must be greater than the minimal distance threshold, α .

2.2 Radial Pursuit

Radial Pursuit (RP) is a novel eye-gaze-based selection method for VR using smooth pursuit, a natural eye movement when our eyes lock onto a moving object. RP can be useful in circumstances where a small target needs to be selected or the target is located among cluttered objects in a small volume where disambiguation is important. Since long dwelling is unnatural for our eyes because they normally saccade several times a second [5, 9], it can be very difficult using only the gaze-dwell technique for selection. To overcome this problem, we leverage smooth pursuit.

Previous research has shown that interaction techniques based on smooth pursuit can be versatile and robust [5]. Nevertheless, we could not find any work applying this technique in immersive VR. RP expands cluttered objects away from each other, reducing the ambiguity and enabling the user to clearly gaze at an object of interest (see Figure 2B-1 to 2B-4). RP can be described in three simple steps:

Step 1: When the user gazes at an area containing an object of interest, a bounding sphere with a predefined radius is created at that location. RP uses the number of objects located within the bounding sphere to calculate the angle to distribute the objects evenly around a circle in the expansion phase.

Step 2: The objects are expanded outward, each object moving continuously until it reaches a defined radius. During this step, the user should gaze at and pursue an object that they would like to select. RP samples the gaze location over the course of the expansion and calculates the Euclidean distance difference between the gaze location and the object's location.

Step 3: The sums of the accumulated distance differences between gaze location, p' , and each object's location, p , are calculated. An object with minimal sum, as shown in Equation 2, is proposed as a candidate for selection. If its sum is found to be below the given confidence threshold, the object is selected.

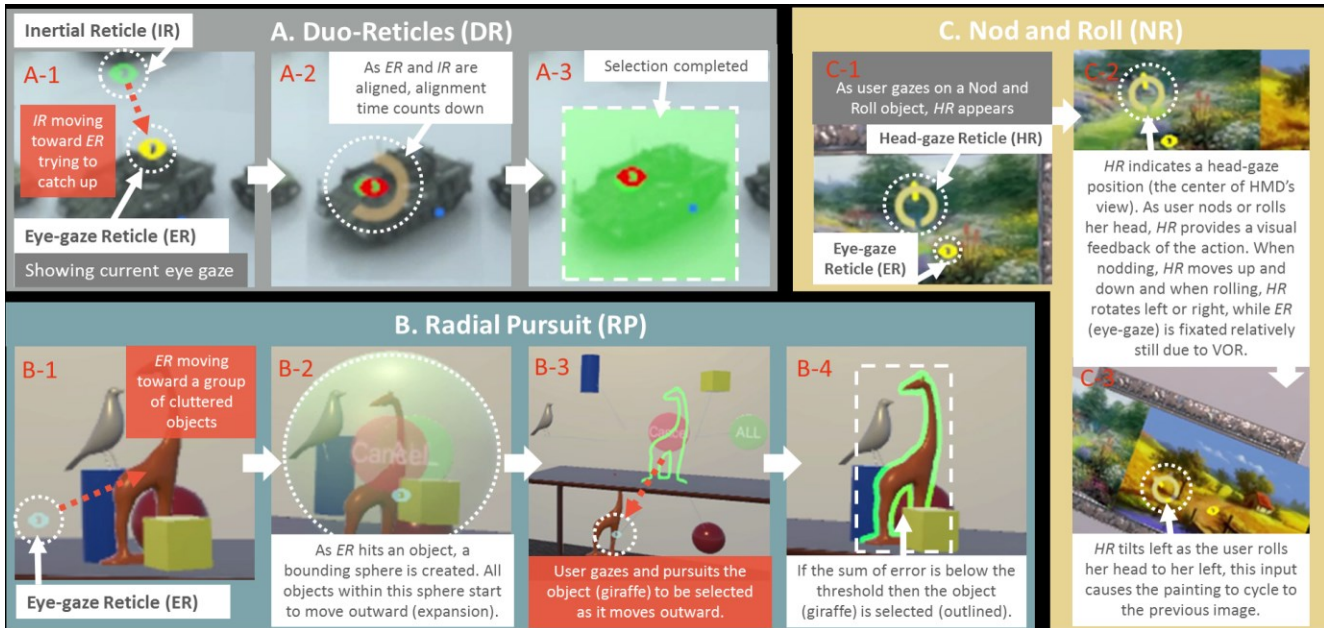


Figure 2: **DR** - (A-1) Green IR moving toward yellow ER, (A-2) Two reticles aligned, orange arc showing align time, (A-3) Selection completed. **RP** - (B-1) ER approaching a group of objects, (B-2) Bounding sphere created, (B-3) Objects expanded, ER pursuing a giraffe, (B-4) A giraffe was selected. **NR** - (C-1) User gazed on a painting, HR was enabled, (C-2) Nodding activated interaction with painting, (C-3) HR tilted left indicating a left head roll, painting rotated.

$$d_{min} = \min(d_0, d_1, \dots, d_k), d_i = \sum_{t=t_{initial}}^n |p(i)_t - p'_t| \quad (2)$$

The time taken for this selection process depends entirely on the expansion time. In our implementation, the expansion time is fixed, however an adaptive expansion time can be used and the pursuit confidence level exceeded much earlier.

2.3 Nod and Roll

Nod and Roll (NR) is a head gesture interaction technique that is designed to take advantage of natural vestibulo-ocular reflex (VOR). Previous research [6-7] has proposed head gestures coupled with eye gaze as an alternate input where VOR provides a natural gaze stabilization. When our eyes fixate on an object, we can move our head freely, because extraocular muscles spontaneously move our eyes in the opposite direction to keep them on the target. We are unaware of any research that considers head gestures in a VR HMD, hence our motivation to explore this design space.

We explore head gestures as discrete inputs for tasks such as browsing a gallery of pictures or controlling a media player. To select an object with NR, users can use head nodding as an input. Eye gaze indicates the object of interest and head nodding is detected based on head acceleration in the vertical direction that is measured using changes in orientation of the HMD. A nod consists of a downward acceleration followed by an upward one exceeding a defined angular threshold within a required span of time.

Head rolls can be mapped to binary inputs for commands like previous and next. Head roll is also simple; if the minimum angular threshold for head roll is exceeded, then a roll is happening (see Figure 2C-1 to 2C-3). If the head returns to the neutral state within a given time, it is considered as a “roll and return.” However, if the head stays longer than a given time, a “roll and stay” is recognized instead. In our implementation, NR can detect seven unique command inputs as follows: (1) nod, (2) roll left and stay, (3) roll right and stay, (4) roll left and return, (5) roll right and return, (6) roll left then right, and (7) roll right then left.

3 USER STUDY

We conducted an initial user study with three parts to test our interaction techniques, where our primary interests were qualitative feedback and usability ratings. The first and second parts tested Duo-Reticles (DR) and Radial Pursuit (RP), respectively, for their performance and usability against a baseline method, Gaze-Dwell (GD). In the third part, participants tried Nod and Roll and gave their impressions. Conditions were carefully balanced in terms of performance and so we predicted comparable performance between our proposed techniques and the baseline. However, we were expecting some differences in usability in favor of our techniques due to our design approach based on natural eye movements.

3.1 Apparatus, Tasks, and Experimental Design

Our hardware equipment included: (1) a laptop PC with an Intel Core i7-6700HQ at 2.60GHz, 16 GB RAM, and NVIDIA GeForce GTX 1070, (2) an iMac with an Intel Core i3 at 3.06 GHz and 4 GB RAM, (3) HTC Vive Kit, (4) a pair of Pupil Labs eye trackers [14] with a binocular mount for the HTC Vive HMD running at 120 Hz. We developed our system using Unity version 5.4.1f1 with the Steam VR plugin for the HTC Vive. The eye tracker was connected to the iMac, which ran Pupil eye tracking software and communicated with the PC over Ethernet. The processing time for eye tracking and network was estimated to be 50-60 ms.

The tasks for each part of the study were as follows:

Part 1: DR vs GD1: The participants had to find a matching picture from sixteen pictures of tanks in a 4x4 grid (see Figure 2A-

1 to 2A-3). GD1 was adjusted for balance in terms of time and error, which yielded a dwell time of 700 ms.

Part 2: RP vs GD2: Participants were asked to select an object out of a cluttered scene (see Figure 2B-1 to 2B-4). GD2 was implemented with radial expansion as in RP, to give a fair comparison, and with dwell time of 700 ms. There was no animation during expansion in GD2, so objects were expanded instantly, and participants had to search and dwell on the object themselves to select.

Part 3: NR: This was a free-interaction session where participants could perform nodding on a wall-hung painting and roll their head to cycle through different paintings (see Figure 2C-1 to 2C-3).

Parts 1 and 2 of this study had a within-subject design with interaction technique being the only independent variable, so there were two conditions in each part. For each condition, participants performed the task with two sets of 11 trials. This resulted in a total of eight sets of trials from both parts of the study.

3.2 Participants and Procedure

Eight volunteers (2 female, ages $m=34.5$, $s.d.=4.21$) participated in the study. All participants had moderate experience in immersive VR ($m=3.5$, $s.d.=1.20$, on a 5-point Likert scale, where 1 was no experience and 5 was expert). The experiment was conducted with participants seated in front of an office desk as shown in Figure 1. After completing a demographic questionnaire, the subject put on the HMD, and the researcher made sure it fit onto the face firmly, and proceeded with a 9-point eye calibration process for the eye tracker. After calibration, participants had a training session for the first part of the study where they were free to try both conditions. Once they were confident with the interaction, the test session began with the order of conditions being counterbalanced. After completing each condition, they were asked to remove the HMD and fill out a questionnaire.

Each time they put the HMD back on, they were asked to align their eye reticle to a blue dot at the center of their view as shown in Figure 2A-1. This dot indicated the head gaze or HMD center viewpoint, and was used for manual alignment between the eye and head tracking without the need for eye recalibration every time the HMD was removed. The study took around 45 minutes to complete. At the end of the study, we conducted a semi-structured interview for feedback on impressions of each technique.

3.3 Results

Table 1: Performance for DR and RP.

	PART 1					PART 2				
	GD1		DR		p	GD2		RP		p
	mean	SD	mean	SD		mean	SD	mean	SD	
total time (s)	92.73	30.68	94.68	22.6	0.38	46.33	17.27	44.75	16.46	0.31
# error	1.92	1.54	1.54	1.28	0.18	2.62	2.79	2.69	2.02	0.46

We used a paired sample t-test to analyze the performance in terms of total time and number of errors between techniques as shown in Table 1. A Wilcoxon signed rank test was used to compare the usability ratings, as shown in Table 2. This analysis was done for Parts 1 and 2.

Part 1: DR vs GD1: There was a significant difference for preference ($W=14$, $p=0.02$) in favor of DR but no significant difference in terms of performance between them.

Part 2: RP vs GD2: There was a significant difference for satisfaction ($W=5$, $p=0.03$) and fatigue ($W=5$, $p=0.03$), both in favor of RP, where RP had higher satisfaction and lower fatigue but no significant difference in performance was found.

Table 2: Results from questionnaire showing 7-points Likert scale usability ratings for each interaction techniques.

	PART 1				PART 2				PART 3			
	Cond	Frequencies	Median	p	Cond	Frequencies	Median	p	Cond	Frequencies	Median	
It felt natural to use	GD1 DR		5 5	0.17	GD2 RP		5 5	0.07	NR		4	
I could interact precisely	GD1 DR		5 4	0.23	GD2 RP		5 5	0.17	NR		4	
It was easy for me to use	GD1 DR		5 5	0.07	GD2 RP		5 5	0.07	NR		5	
I felt satisfied using it	GD1 DR		5 5	0.14	GD2 RP		5 6	0.03	NR		5	
It was fun to use	GD1 DR		6 6	0.14	GD2 RP		6 6	0.07	NR		6	
I need to concentrate to use it	GD1 DR		6 6	0.33	GD2 RP		6 6	0.09	NR		5	
I felt tired using it	GD1 DR		5 6	0.09	GD2 RP		5 4	0.03	NR		4	
It was frustrating to use	GD1 DR		3 2	0.14	GD2 RP		2 3	0.30	NR		3	
I preferred this technique	GD1 DR		2 6	0.02	GD2 RP		3 5	0.12				

1 2 3 4 5 6 7

Part 3: NR: The usability ratings for NR were generally rated positively, with the strongest attribute being that it was fun to use.

3.4 Discussion

Although our initial study had a small sample size, which reduced the statistical power, we could find significant differences in terms of subjective ratings in favor of our methods. We could also confirm that there was no significant performance difference between our methods and the baseline as we carefully balanced each condition to create a fair test. This early finding is a positive indication that designing eye-gaze-based interaction around natural eye movements could improve the user experience while maintaining comparable performance to standard interaction techniques.

We also obtained interesting feedback for each technique from the semi-structured interviews:

DR: Participants felt the interaction was almost implicit as the alignment time was short compared to GD1. Although there was no difference in performance, they did not notice the travelling time required by the IR as they were busy looking for the right match. They also felt they had more control with DR. P05 stated: "I felt time pressure with GD1. With DR, I felt I had time to look and I knew where the other reticle was." However, some participants were briefly distracted by the IR as it moved toward their gaze location and accidentally gazed at it. Eventually, they became accustomed to the second reticle and could understand its behavior. As P01 pointed out, "I accidentally looked at the green reticle when it came close, but I got used to it." We believe that the issue could be addressed by having an adaptive reticle that changes color depending on the background, so that it could still be seen, but does not distract the user.

RP: Again, participants did not perceive the waiting time during their pursuit of the moving object as opposed to dwelling on the object. P02 stated that, "RP was easy, just follow the object and it was selected." However, some participants preferred to take their time to look for the right object. As P01 pointed out, "With RP, I need to know in advance the object to follow, but with GD2, I could wait until it expanded and look for the object I wanted to select." Another problem was that if the participant pursued too late, the selection would be cancelled, as the confidence level was too low. We believe both issues could be addressed by using a fallback method when the confidence level is too low after the RP period by using GD with a shorter dwell time of 200-300 ms.

NR: Participants found it amusing to use their head for interaction. Most participants found it fun and engaging. Some felt that it was alright if they did not need to gesture all the time. We found that strong head movement could lead to shifts in the HMD's

position on the face, and this could invalidate the eye calibration. P07 expressed this, "The HMD was not very light, so it was awkward to gesture, especially nodding". We expect head gestures to be good for input as HMDs become lighter, and better methods for securing the HMD on the user's face are found.

4 CONCLUSION AND FUTURE WORK

We presented three novel eye-gaze-based interaction techniques inspired by natural eye movements. An initial study found positive results supporting our approaches. Our techniques had similar performance with Gaze-Dwell, but superior user experience. We plan to conduct a follow-up study with a larger sample size and more dependent variables. We will continue to apply the same principles in design to improve user experience using eye gaze for immersive VR.

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